

Final Project

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Load and Clean Data

```
GliomaData <- read.csv('GliomaData/TCGA_GBM_LGG_Mutations_all.csv')
head(GliomaData)
```

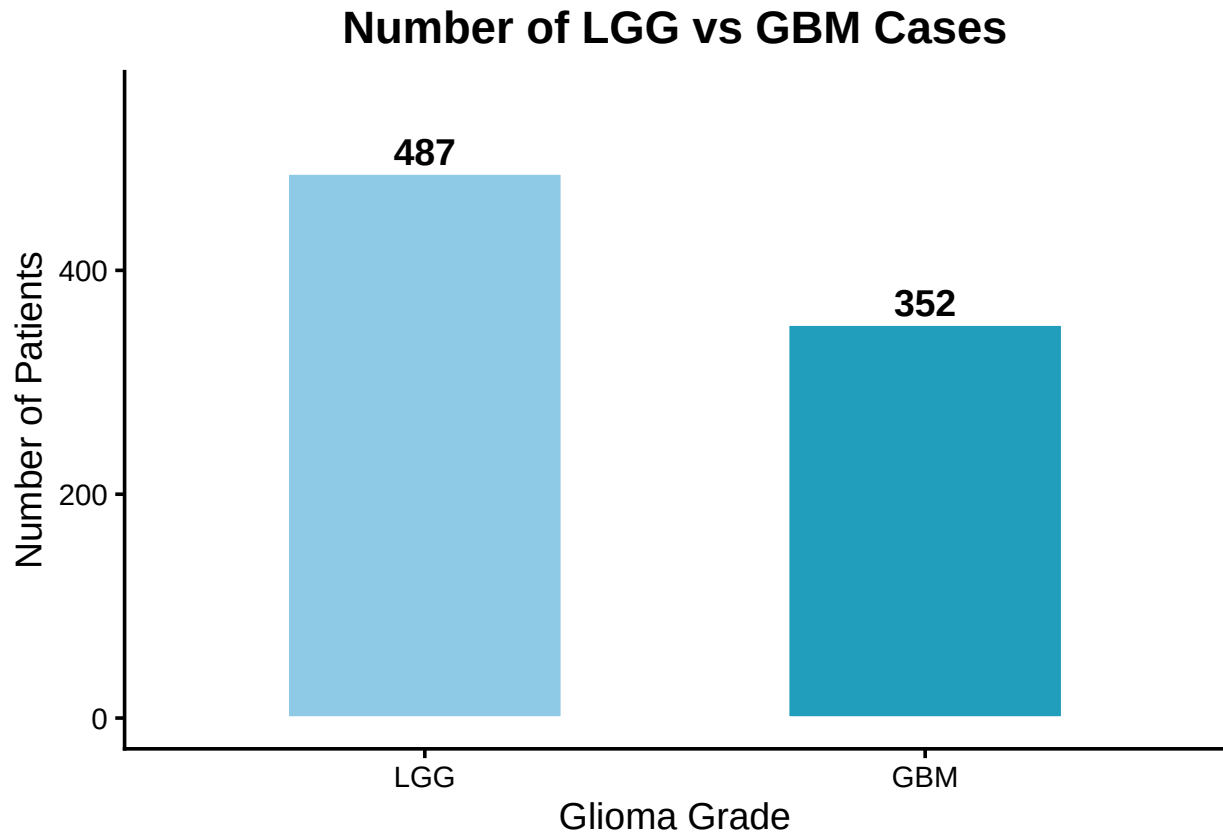
```
GliomaINFO <- read.csv('GliomaData/TCGA_InfoWithGrade.csv')
head(GliomaINFO)
```

```
unique(GliomaData[, c("Grade", "Primary_Diagnosis")])
```

```
##      Grade      Primary_Diagnosis
## 1     LGG      Oligodendroglioma, NOS
## 2     LGG      Mixed glioma
## 3     LGG      Astrocytoma, NOS
## 4     LGG      Astrocytoma, anaplastic
## 7     LGG      Oligodendroglioma, anaplastic
## 42    LGG      --
## 500   GBM      Glioblastoma
## 672   GBM      --
```

There are no missing data.

```
GliomaINFO %>%
  count(Grade) %>%
  mutate(Grade = factor(Grade, levels = c(0, 1),
                        labels = c("LGG", "GBM"))) %>%
  ggplot(aes(x = Grade, y = n, fill = Grade)) +
  geom_col(width = 0.55, color = "white") +
  geom_text(aes(label = n),
            vjust = -0.3, size = 5, fontface = "bold") +
  scale_fill_manual(values = c("#8ecae6", "#219ebc")) +
  scale_y_continuous(limits = c(0, 550)) +
  labs(
    title = "Number of LGG vs GBM Cases",
    x = "Glioma Grade",
    y = "Number of Patients"
  ) +
  theme_classic(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```



LGG cases are more frequent than GBM cases in our sample, which may influence class imbalance for downstream modeling.

```
GliomaINFO %>%
  mutate(Grade = factor(Grade, labels = c("LGG", "GBM"))) %>%
  ggplot(aes(x = Grade, y = Age_at_diagnosis, fill = Grade)) +
  geom_boxplot(width = 0.6, alpha = 0.85) +
  scale_fill_manual(values = c("#dda15e", "#bc6c25")) +
  labs(
    title = "Age Distribution by Glioma Grade",
    x = "Glioma Grade",
    y = "Age at Diagnosis (years)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```



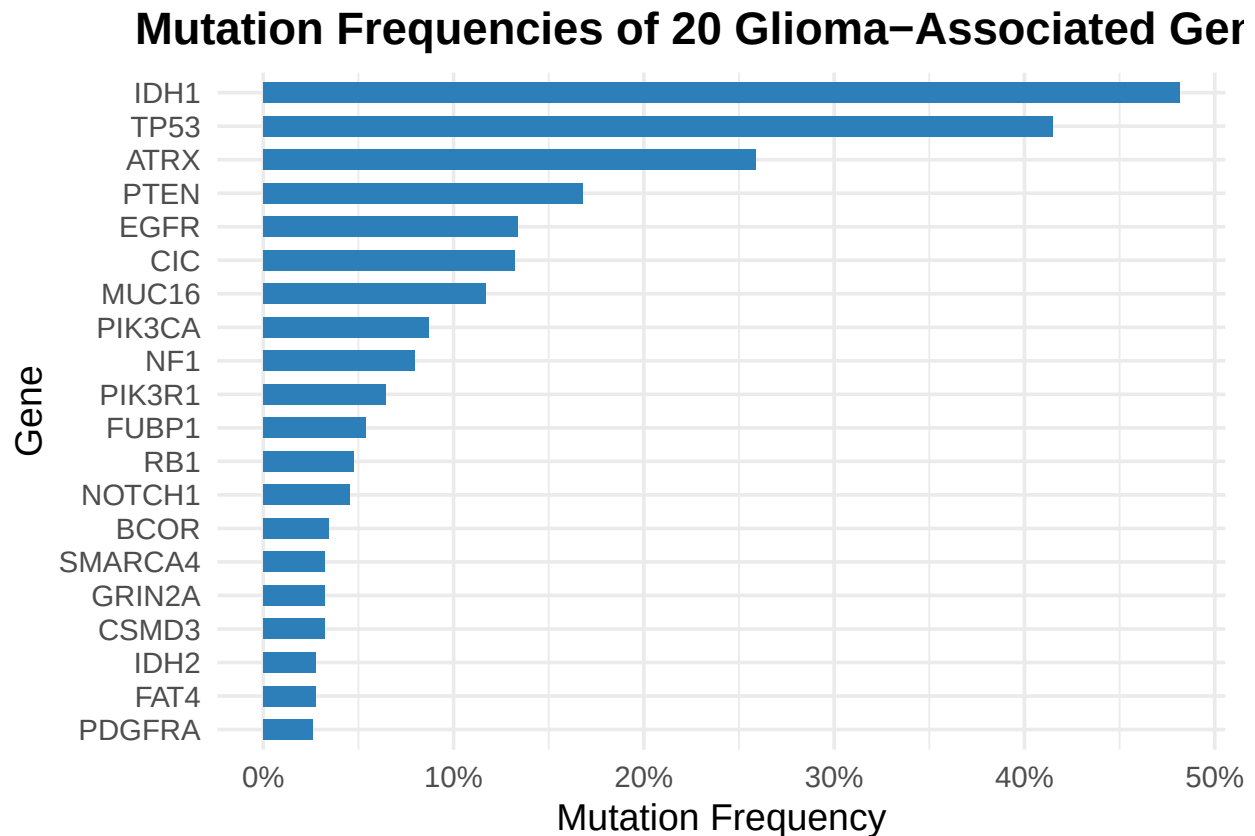
This boxplot shows the age distribution for LGG and GBM patients. We can see that LGG patients are generally younger, with a median age in the early 40s. GBM patients are older on average, with a median close to 60. This matches what we usually see in glioma cases and shows a clear age difference between the two groups before we look at genetic mutations.

```
mut_long <- GliomaINFO %>%
  select(IDH1:PDGFRA) %>%
  pivot_longer(cols = everything(),
               names_to = "Gene",
               values_to = "Mutated")

mut_freq <- mut_long %>%
  group_by(Gene) %>%
  summarize(freq = mean(Mutated), .groups = "drop")

ggplot(mut_freq, aes(x = reorder(Gene, freq), y = freq)) +
  geom_col(fill = "#2c7fb8", width = 0.6) +
  coord_flip() +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Mutation Frequencies of 20 Glioma-Associated Genes",
    x = "Gene",
    y = "Mutation Frequency"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5)
```

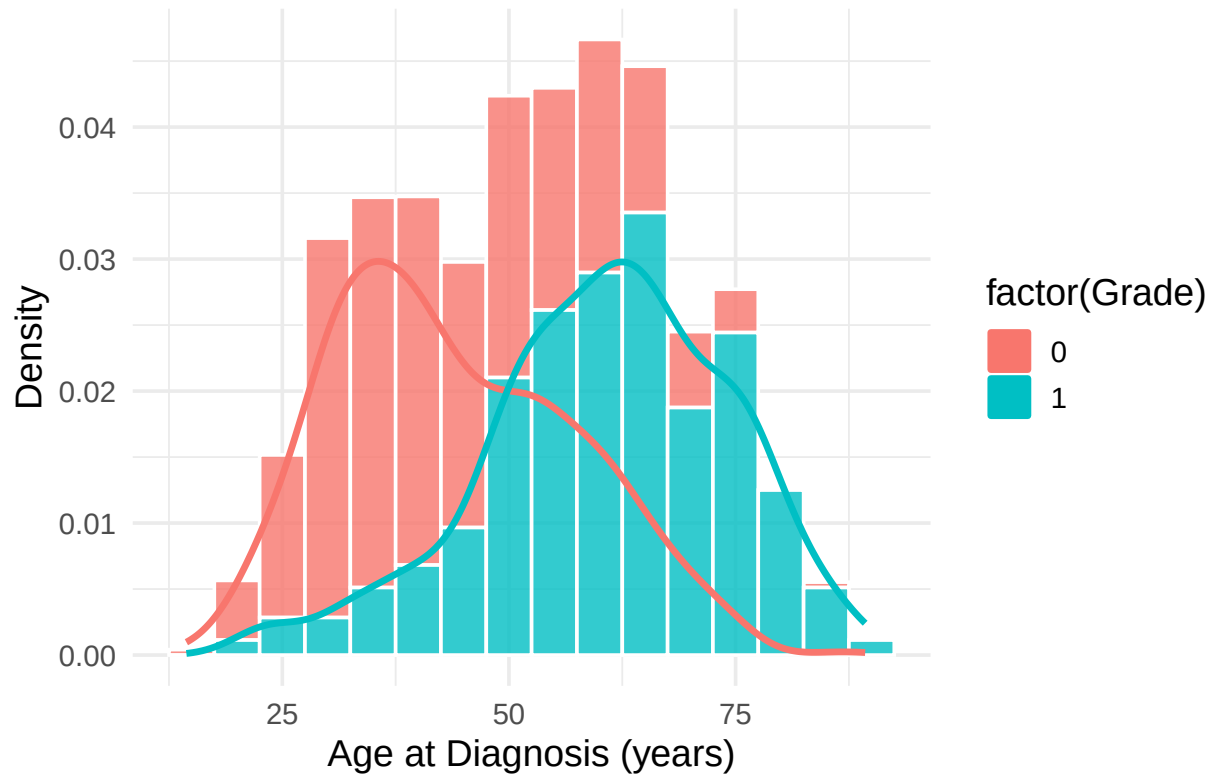
)



This figure shows the mutation frequency of 20 glioma-associated genes. IDH1, TP53, and ATRX are the most frequently mutated genes in our dataset, which matches what is usually seen in glioma biology. Many other genes show much lower mutation rates, which also explains why some chi-square tests required Fisher's exact test due to small expected counts.

```
GliomaINFO %>%
  ggplot(aes(x = Age_at_diagnosis)) +
  geom_histogram(aes(y = after_stat(density), fill = factor(Grade))
    , binwidth = 5, color = "white", alpha = 0.8) +
  geom_density(
    aes(color = factor(Grade)), # also color by Grade
    linewidth = 1.2
  ) +
  labs(
    title = "Age Distribution with Density Curve",
    x = "Age at Diagnosis (years)",
    y = "Density"
  ) +
  theme_minimal(base_size = 14)
```

Age Distribution with Density Curve



```
shapiro.test(GliomaINFO$Age_at_diagnosis)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  GliomaINFO$Age_at_diagnosis
## W = 0.98368, p-value = 4.653e-08
```

The Shapiro-Wilk test ($p < 0.001$) indicates that age at diagnosis is not normally distributed. This matches the density plot, which shows a right-skewed and multimodal distribution due to younger LGG patients and older GBM patients.

ESTABLISH COLINEARITY

```
threshold <- 0.5
corr_mat <- cor(GliomaINFO, use = "pairwise.complete.obs")

high_corr_pairs <- which(abs(corr_mat) > threshold & lower.tri(corr_mat),
                        , arr.ind = TRUE)

pairs_df <- data.frame(
  Var1 = rownames(corr_mat)[high_corr_pairs[,1]],
  Var2 = colnames(corr_mat)[high_corr_pairs[,2]],
  Correlation = corr_mat[high_corr_pairs]
)

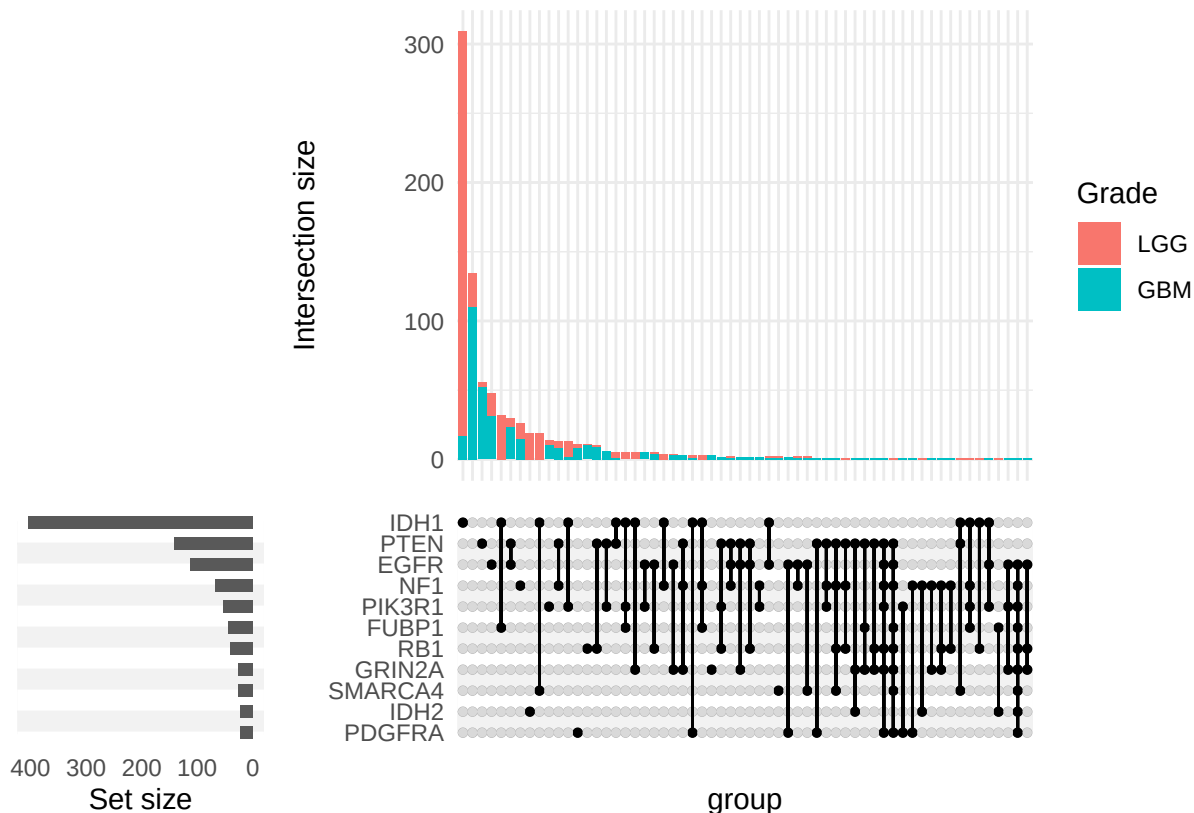
pairs_df
```

```
##           Var1           Var2 Correlation
## 1 Age_at_diagnosis      Grade  0.5292029
## 2           IDH1      Grade -0.7081408
## 3           IDH1 Age_at_diagnosis -0.5688922
## 4           ATRX      TP53   0.5469031

library(ComplexUpset)
library(ggplot2)

mut_cols <- c("IDH1", "PTEN", "EGFR", "NF1"
              , "PIK3R1", "FUBP1", "RB1", "SMARCA4"
              , "GRIN2A", "IDH2", "PDGFRA") # replace with your columns
GliomaINFO$Grade <- factor(
  GliomaINFO$Grade,
  levels = c("0", "1"),
  labels = c("LGG", "GBM")
)

GliomaINFO %>%
  upset(
    mut_cols,
    base_annotations = list(
      'Intersection size' = (
        intersection_size(
          counts = FALSE,
          mapping = aes(fill = Grade)
        )
      )
    ),
    matrix = intersection_matrix(
      geom = geom_point(size = 1)
    ),
    themes = upset_modify_themes(
      theme(
        panel.spacing = unit(14, "pt"),
        axis.text.x = element_text(angle = 45, hjust = 1, size = 5),
        plot.margin = margin(t = -10, b = 10)
      )
    )
  )
)
```



More of the cases with just an IDH1 mutation have LGG, while the groups having no mutation or just PTEN or just EGFR have more GBM diagnosis.

```
tbl_summary(GliomaData %>% select(-c(Case_ID, Age_at_diagnosis, Project
, Primary_Diagnosis)), by = Grade)
```

```
# Add p-value to tbl
```

Because ATRX has a stronger correlation with Grade than TP53 would, we will remove TP53 from the model.

```
muts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

for (m in muts) {
  cat("\n==== Testing:", m, "====\n")

  tab <- table(GliomaData$Grade, GliomaData[[m]])
  print(tab)

  exp_counts <- chisq.test(tab)$expected
  cat("\nExpected counts:\n")
  print(round(exp_counts, 2))

  # Check assumption
  if (any(exp_counts < 1) || mean(exp_counts < 5) > 0.2) {
    cat("\n Chi-square assumptions NOT met -> using Fisher's exact test.\n")
    print(fisher.test(tab))
  }
}
```

```

} else {
  cat("\n Chi-square assumptions OK.\n")
  print(chisq.test(tab))
}
}

```

```

##
## ===== Testing: IDH1 =====
##
##      MUTATED NOT_MUTATED
##  GBM      23      340
##  LGG     391     108
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM  174.34   188.66
##  LGG  239.66   259.34
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 433.79, df = 1, p-value < 2.2e-16
##
##
## ===== Testing: TP53 =====
##
##      MUTATED NOT_MUTATED
##  GBM      116      247
##  LGG      238      261
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM  149.07   213.93
##  LGG  204.93   294.07
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 20.864, df = 1, p-value = 4.931e-06
##
##
## ===== Testing: ATRX =====
##
##      MUTATED NOT_MUTATED
##  GBM      35      328
##  LGG     185      314
##
## Expected counts:

```



```

##
##      MUTATED NOT_MUTATED
##   GBM   92.65     270.35
##   LGG  127.35     371.65
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 81.755, df = 1, p-value < 2.2e-16
##
##
## ===== Testing: PTEN =====
##
##      MUTATED NOT_MUTATED
##   GBM     119      244
##   LGG     25      474
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##   GBM   60.64     302.36
##   LGG   83.36     415.64
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 114.49, df = 1, p-value < 2.2e-16
##
##
## ===== Testing: EGFR =====
##
##      MUTATED NOT_MUTATED
##   GBM      83      280
##   LGG      31      468
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##   GBM   48.01     314.99
##   LGG   65.99     433.01
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 49.337, df = 1, p-value = 2.156e-12
##
##
## ===== Testing: CIC =====

```

```

##
##      MUTATED NOT_MUTATED
##   GBM      4      359
##   LGG     110     389
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##   GBM  48.01   314.99
##   LGG  65.99   433.01
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 78.492, df = 1, p-value < 2.2e-16
##
##
## ===== Testing: MUC16 =====
##
##      MUTATED NOT_MUTATED
##   GBM      59      304
##   LGG      41      458
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##   GBM  42.11   320.89
##   LGG  57.89   441.11
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 12.464, df = 1, p-value = 0.0004149
##
##
## ===== Testing: PIK3CA =====
##
##      MUTATED NOT_MUTATED
##   GBM      36      327
##   LGG      41      458
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##   GBM  32.43   330.57
##   LGG  44.57   454.43
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction

```

```

##
## data:  tab
## X-squared = 0.55288, df = 1, p-value = 0.4571
##
##
## ===== Testing: NF1 =====
##
##      MUTATED NOT_MUTATED
##  GBM      40      323
##  LGG      29      470
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM  29.06   333.94
##  LGG  39.94   459.06
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 7.0478, df = 1, p-value = 0.007936
##
##
## ===== Testing: PIK3R1 =====
##
##      MUTATED NOT_MUTATED
##  GBM      35      328
##  LGG      22      477
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM      24      339
##  LGG      33      466
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 8.4905, df = 1, p-value = 0.00357
##
##
## ===== Testing: FUBP1 =====
##
##      MUTATED NOT_MUTATED
##  GBM       2      361
##  LGG      45      454
##
## Expected counts:
##
##      MUTATED NOT_MUTATED

```

```

##   GBM   19.79   343.21
##   LGG   27.21   471.79
##
##   Chi-square assumptions OK.
##
##   Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 27.604, df = 1, p-value = 1.489e-07
##
##
## ===== Testing: RB1 =====
##
##           MUTATED NOT_MUTATED
##   GBM         35         328
##   LGG          6         493
##
## Expected counts:
##
##           MUTATED NOT_MUTATED
##   GBM    17.27    345.73
##   LGG    23.73    475.27
##
##   Chi-square assumptions OK.
##
##   Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 31.202, df = 1, p-value = 2.326e-08
##
##
## ===== Testing: NOTCH1 =====
##
##           MUTATED NOT_MUTATED
##   GBM          0         363
##   LGG         38         461
##
## Expected counts:
##
##           MUTATED NOT_MUTATED
##   GBM         16         347
##   LGG         22         477
##
##   Chi-square assumptions OK.
##
##   Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 27.139, df = 1, p-value = 1.893e-07
##
##
## ===== Testing: BCOR =====
##
##           MUTATED NOT_MUTATED

```

```

##      GBM      12      351
##      LGG      17      482
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##      GBM    12.21    350.79
##      LGG    16.79    482.21
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 8.7483e-29, df = 1, p-value = 1
##
##
## ===== Testing: CSMD3 =====
##
##      MUTATED NOT_MUTATED
##      GBM      15      348
##      LGG      13      486
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##      GBM    11.79    351.21
##      LGG    16.21    482.79
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 1.1111, df = 1, p-value = 0.2918
##
##
## ===== Testing: SMARCA4 =====
##
##      MUTATED NOT_MUTATED
##      GBM       4      359
##      LGG      24      475
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##      GBM    11.79    351.21
##      LGG    16.21    482.79
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab

```

```

## X-squared = 8.0498, df = 1, p-value = 0.004551
##
##
## ===== Testing: GRIN2A =====
##
##      MUTATED NOT_MUTATED
##  GBM      20      343
##  LGG       7      492
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM   11.37   351.63
##  LGG   15.63   483.37
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 10.367, df = 1, p-value = 0.001283
##
##
## ===== Testing: IDH2 =====
##
##      MUTATED NOT_MUTATED
##  GBM       2      361
##  LGG      21      478
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM    9.69   353.31
##  LGG   13.31   485.69
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 9.4613, df = 1, p-value = 0.002098
##
##
## ===== Testing: FAT4 =====
##
##      MUTATED NOT_MUTATED
##  GBM      12      351
##  LGG      11      488
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
##  GBM    9.69   353.31
##  LGG   13.31   485.69

```

```
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 0.60323, df = 1, p-value = 0.4373
##
##
## ===== Testing: PDGFRA =====
##
##      MUTATED NOT_MUTATED
## GBM      16      347
## LGG       6      493
##
## Expected counts:
##
##      MUTATED NOT_MUTATED
## GBM    9.26    353.74
## LGG   12.74    486.26
##
## Chi-square assumptions OK.
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 7.4397, df = 1, p-value = 0.00638
```

The following mutations do not show a significant difference between the two Glioma grades: PIK3CA, BCOR, CSMD3, FAT4 We can omit them from the clustering model.

Logistic Regression Model

```
removee <- c('PIK3CA', 'BCOR', 'CSMD3', 'FAT4', 'TP53'
             , 'Gender', 'Race', 'ATRX', 'CIC', 'MUC16', 'NOTCH1')

mData <- GliomaINFO %>%
  select(-all_of(removee))
```

```
model <- glm(Grade ~ .,
             data = mData,
             family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Grade ~ ., family = binomial, data = mData)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.663408   0.528727  -1.255 0.209578
## Age_at_diagnosis  0.033421   0.008627   3.874 0.000107 ***
## IDH1            -3.595340   0.304411 -11.811 < 2e-16 ***
## PTEN             0.708951   0.290994   2.436 0.014838 *
## EGFR            -0.614537   0.285133  -2.155 0.031141 *
```

```

## NF1          -1.038327   0.340982  -3.045 0.002326 **
## PIK3R1       1.084272   0.503084   2.155 0.031142 *
## FUBP1       -1.883979   1.095646  -1.720 0.085521 .
## RB1         0.824066   0.550947   1.496 0.134725
## SMARCA4     -0.678433   0.731794  -0.927 0.353884
## GRIN2A      1.757979   0.700572   2.509 0.012095 *
## IDH2       -3.262513   0.827602  -3.942 8.08e-05 ***
## PDGFRA      0.692521   0.728625   0.950 0.341884
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1141.28  on 838  degrees of freedom
## Residual deviance:  557.03  on 826  degrees of freedom
## AIC: 583.03
##
## Number of Fisher Scoring iterations: 6

```


Characteristic	GBM N = 363 ¹	LGG N = 499 ¹
Gender		
–	3 (0.8%)	1 (0.2%)
Female	138 (38%)	221 (44%)
Male	222 (61%)	277 (56%)
Race		
–	3 (0.8%)	1 (0.2%)
american indian or alaska native	0 (0%)	1 (0.2%)
asian	6 (1.7%)	8 (1.6%)
black or african american	38 (10%)	21 (4.2%)
not reported	8 (2.2%)	10 (2.0%)
white	308 (85%)	458 (92%)
IDH1		
MUTATED	23 (6.3%)	391 (78%)
NOT_MUTATED	340 (94%)	108 (22%)
TP53		
MUTATED	116 (32%)	238 (48%)
NOT_MUTATED	247 (68%)	261 (52%)
ATRX		
MUTATED	35 (9.6%)	185 (37%)
NOT_MUTATED	328 (90%)	314 (63%)
PTEN		
MUTATED	119 (33%)	25 (5.0%)
NOT_MUTATED	244 (67%)	474 (95%)
EGFR		
MUTATED	83 (23%)	31 (6.2%)
NOT_MUTATED	280 (77%)	468 (94%)
CIC		
MUTATED	4 (1.1%)	110 (22%)
NOT_MUTATED	359 (99%)	389 (78%)
MUC16		
MUTATED	59 (16%)	41 (8.2%)
NOT_MUTATED	304 (84%)	458 (92%)
PIK3CA		
MUTATED	36 (9.9%)	41 (8.2%)
NOT_MUTATED	327 (90%)	458 (92%)
NF1		
MUTATED	40 (11%)	29 (5.8%)
NOT_MUTATED	323 (89%)	470 (94%)
PIK3R1		
MUTATED	35 (9.6%)	22 (4.4%)
NOT_MUTATED	328 (90%)	477 (96%)
FUBP1		
MUTATED	2 (0.6%)	45 (9.0%)
NOT_MUTATED	361 (99%)	454 (91%)
RB1		
MUTATED	35 (9.6%)	6 (1.2%)
NOT_MUTATED	328 (90%)	493 (99%)
NOTCH1		
MUTATED	0 (0%)	38 (7.6%)
NOT_MUTATED	363 (100%)	461 (92%)